SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

ASSESSMENT TOOL 2 – Coding Golf Competition

Course Code:	CSA03	**Course Title:**	Data Structures
CO Assessed:	CO4 –Articulation (BL3)	**Assessment:**	Assessment Tool 2 – Coding Golf Competition
Weightage:	20% of CIA	**Total Marks:**	50
CO3 Statement:	Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data		
Student Name:	suriya priya k	**Reg. No.:**	192521195
Section / Batch:		**Date:**	12/02026

Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions

- This test contains 5 Coding challenge Total: 50 Marks.
- Evaluation of Answer Quality
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Hashing - Priority Queue & Heaps-	Hashing, Heap	1,2	20
Sorting Algorithm	Insertion sort, Heap sort	3,4	20
Parallel Sorting algorithms-Time complexity.	Merge Sort, Quick sort	5	10
		GRAND TOTAL:	50Marks

Annexure A – Coding Golf Competition

Q1) Linear Probing – Basic
Given the keys:
23, 43, 13, 27, 18, 33, 63
and hash function:
Write a program to insert all keys into a hash table using linear probing. Display the final hash table.
Golf condition: Minimum number of lines of correct code.
Q2) Separate Chaining
Implement a hash table of size 10 using separate chaining for:
25, 35, 15, 42, 52, 62, 72
using:
Display the resulting chains.
Golf condition: Produce the shortest correct implementation.
Q3) Search Using Hashing
Given:
18, 41, 22, 44, 59, 32, 31
construct a hash table using linear probing and write a program to search for a given key.
Challenge: The program should report the number of probes required.
Q4) Collision Counter
Write a program that inserts N keys into a hash table using linear probing and counts the total number of collisions.
Example:
Input:
5
23 43 13 27 33
Output:
Collisions = ?
Golf: Shortest correct code wins.

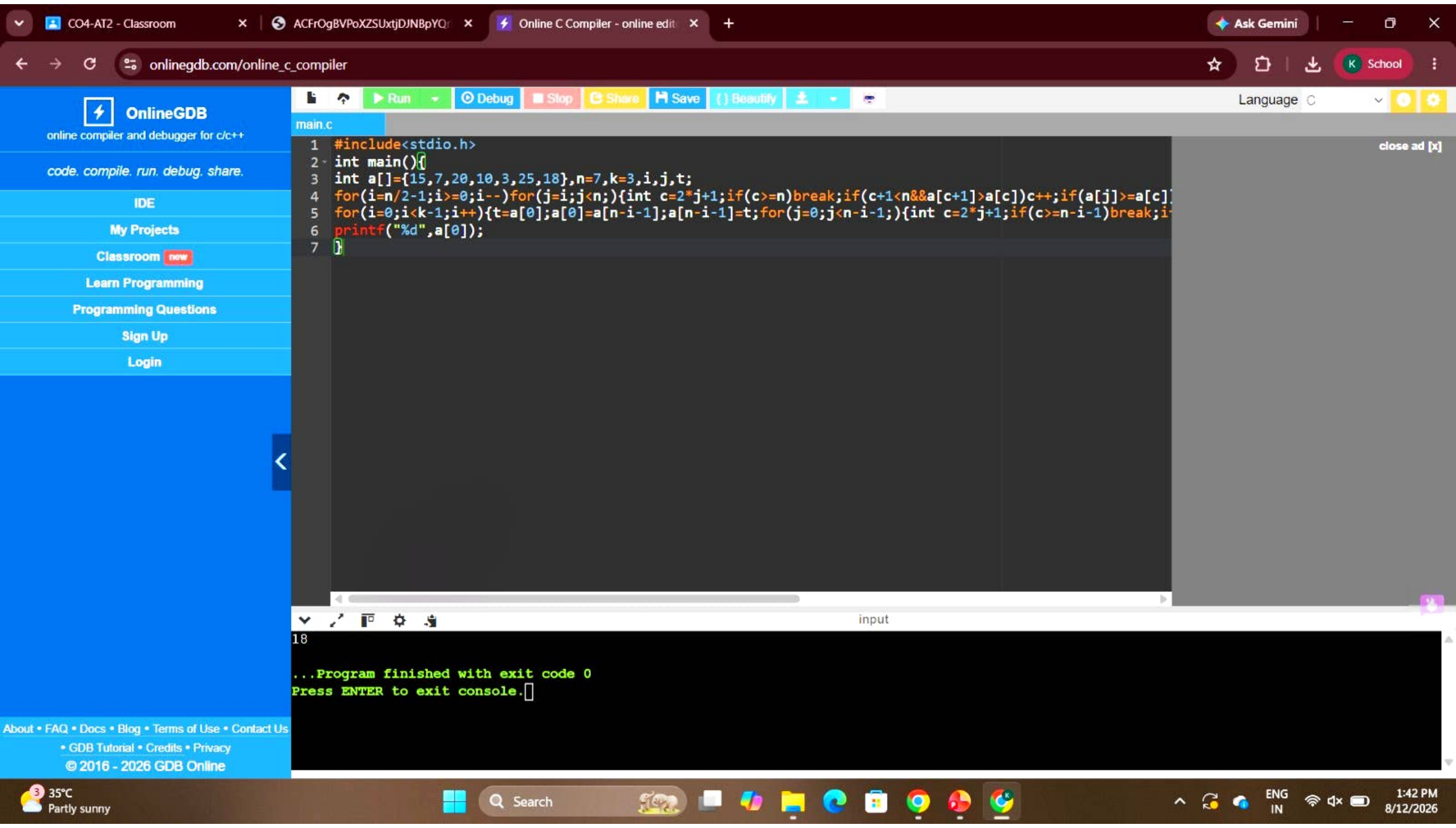
Q5) Q5. Kth Largest Element
Given:
15 7 20 10 3 25 18
and K = 3, find the 3rd largest element using a heap.
Expected output:
18
Golf condition: Do not use a built-in sorting function.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct output	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Algorithm Implementation	25%	Provides detailed and comprehensive comparison	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison

		across multiple aspects			
Code optimization	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Time complexity	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningf ul interpretat ion
Code readability	15%	Well-organize d, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________



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Language C
main.c
close ad [x]
#include<stdio.h>
int main(){
int a[]={15,7,20,10,3,25,18},n=7,k=3,i,j,t;
for(i=n/2-1;i>=0;i--)for(j=i;j<n;){int c=2*j+1;if(c>=n)break;if(c+1<n&&a[c+1]>a[c])c++;if(a[j]>=a[c]
for(i=0;i<k-1;i++){t=a[0];a[0]=a[n-i-1];a[n-i-1]=t;for(j=0;j<n-i-1;){int c=2*j+1;if(c>=n-i-1)break;i
printf("%d",a[0]);
}
input
18
...Program finished with exit code 0
Press ENTER to exit console.
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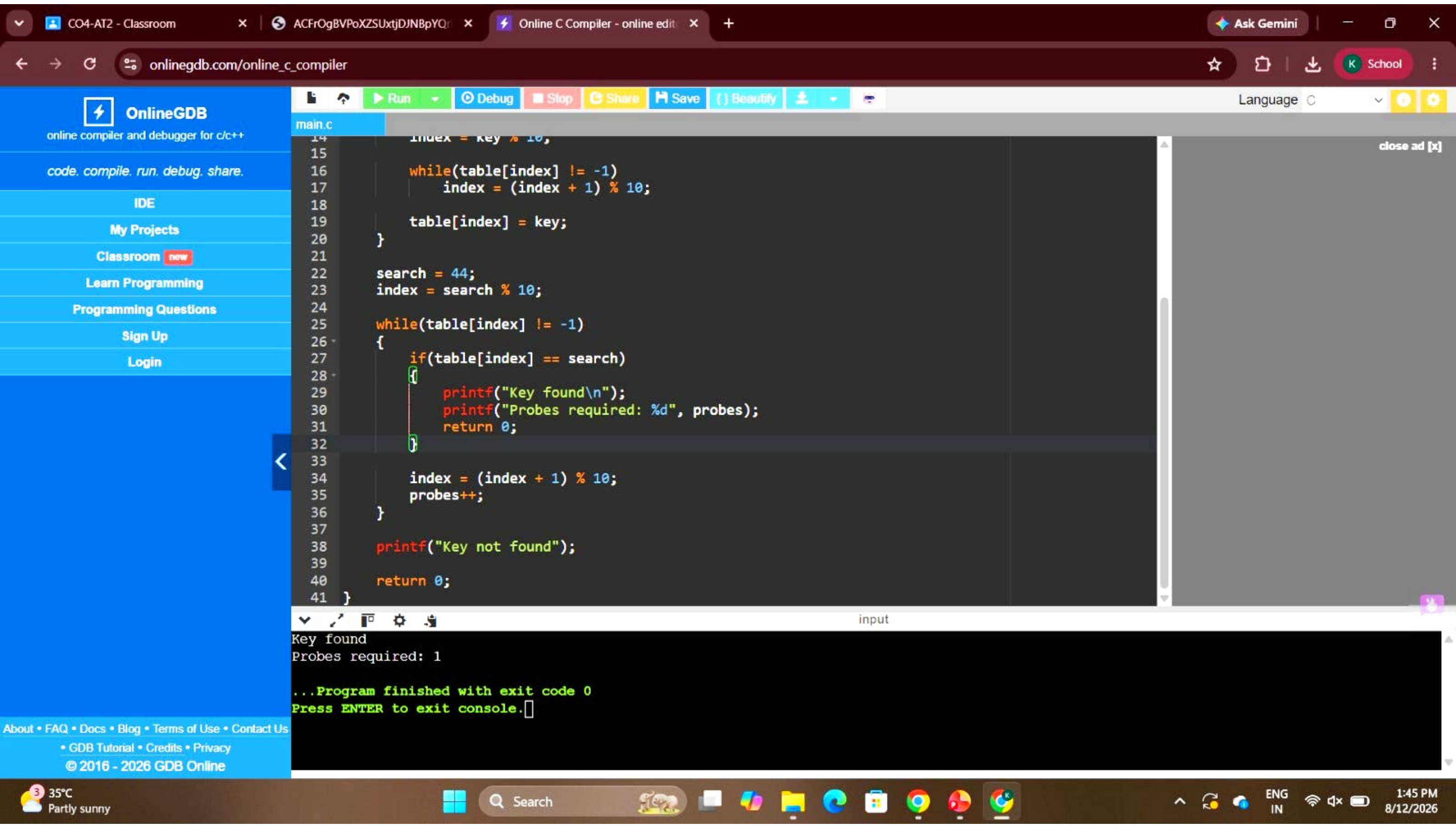
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Language C
main.c
close ad [x]
    while(table[index] != -1)
        index = (index + 1) % 10;
    table[index] = key;
}
search = 44;
index = search % 10;
while(table[index] != -1)
{
    if(table[index] == search)
    {
        printf("Key found\n");
        printf("Probes required: %d", probes);
        return 0;
    }
    index = (index + 1) % 10;
    probes++;
}
printf("Key not found");
return 0;
}
input
Key found
Probes required: 1
...Program finished with exit code 0
Press ENTER to exit console.
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Language C

main.c

```c
    int table[10][10] = {0};
    int count[10] = {0};
    int keys[7] = {25, 35, 15, 42, 52, 62, 72};
    int i, key, index;

    for(i = 0; i < 7; i++)
    {
        key = keys[i];
        index = key % 10;

        table[index][count[index]] = key;
        count[index]++;
    }

    printf("Hash Table:\n");

    for(i = 0; i < 10; i++)
    {
        printf("%d : ", i);

        for(int j = 0; j < count[i]; j++)
            printf("%d ", table[i][j]);

        printf("\n");
    }

    return 0;
}
```

close ad [x]

input

```
Hash Table:
0 :
1 :
2 : 42 52 62 72
3 :
4 :
5 : 25 35 15
6 :
```


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main.c

```c
#include <stdio.h>

int main()
{
    int table[10], keys[7] = {23, 43, 13, 27, 18, 33, 63};
    int i, key, index;

    for(i = 0; i < 10; i++)
        table[i] = -1;

    for(i = 0; i < 7; i++)
    {
        key = keys[i];
        index = key % 10;

        while(table[index] != -1)
            index = (index + 1) % 10;

        table[index] = key;
    }

    printf("Final Hash Table:\n");

    for(i = 0; i < 10; i++)
        printf("%d : %d\n", i, table[i]);

    return 0;
```

```
6 : 33
7 : 27
8 : 18
9 : 63


...Program finished with exit code 0
Press ENTER to exit console.
```